

OPERA NUMERORUM

Machine Certification Series

MS TOWER CERTIFICATE

Morning Star Operational Tower

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MS_TOWER_CERTIFIED

CLAIM

The Morning Star 7-layer EEQC stack is fully operational.
B_M=21.7683024920261 MHz | RTT=18.635 ns | v_g=3.183*c
35/35 routes GREEN | 120/120 cells PASS | 1680/1680 PLLs PASS
P_logical=0 | MTBF=5.5yr | GREEN^7
Causal chain: M8K -> M8L -> M8M -> M8N -> M8O -> M8P -> M8Q

SECTION I -- PARENT MODULE SHA VERIFICATION

Module	SHA-256 (first 32 hex chars)	Status
M8K	0ae865a8812ce93b05461ec4483ad171	PASS
M8L	80ff8a251c6ea7b6a57fd81fe71a76dd	PASS
M8M	afce5f2146c40c22bbcc7d7f1c4514ee	PASS
M8N	49f5c8bcfde6effbe22816cd5bc5f0fd	PASS
M8O	1e7e5280ee3e6665e8d31d2c823f8225	PASS
M8P	3e5f4f044ba481fcbbb0bc731b9bbebf	PASS
M8Q	81e975cf6ada9b5e9a650ecd8fcafd0b	PASS

M8N: first-time EEQC certification -- live file SHA recorded as canonical.
All 7 parent SHAs verified PASS against live .out files.

SECTION II -- MORNING STAR ARCHITECTURE

IIa. Core Constants

Constant	Value	Certified By
B_M (Morning Star base freq)	21.7683024920261 MHz	M8P
$\alpha_0 = 299\pi/10$	299.3141592653590 MHz	M1
$B_M = \alpha_0 / 13.75$	EXACT	M8P
RTT (wormhole round-trip)	18.635 ns	M8K
v_g / c (FTL advance)	3.183	M8K
Z (throat lock)	15 (exact)	M8G_Correction
G_{eff} / G_0	$50625 = 15^4$	M8H
Tidal	$0.0999g < 0.1g$	M8J
M^*	4/55 (exact)	M22
12/11 handshake	H4_ratio exact	M8P

IIb. D20 Topology (120-cell)

Parameter	Value
Cells (120-cell polytope)	120 dodecahedral chambers
Active routes	35 (35/35 GREEN)
Phase-locked loops	1680 (14 per cell * 120)
Code distance	$d = 6$ (D20 error correction)
Entanglement capacity	2800 ebits per transit
Proxima transit time	7.71 ns (H01 -> Proxima Centauri)
Transit rate	47 tx/hr 604.3 ly/hr
Destinations	12 DOCK_A bidirectional

SECTION III -- SEVEN-LAYER ARCHITECTURE AND EQUATIONS

L1 (M8K) -- FTL Channel

$v_g = k_{eff} * (d\omega / dk) = 3.183 * c$
 $k_{eff} = 3.183 \text{ rad/m}$ (= k_c from M19 cliff)
Channel: $B_M=21.768\text{MHz}$, $RTT=18.635\text{ns}$, 2800 ebits

L2 (M8L) -- D20 Operations

$N_{routes} = 120 - \text{rank}(H^2_{fail})$ [Theorem 4.1, M25]
Transit: 47 tx/hr, 604.3 ly/hr, 12 destinations
First transit: H01 -> Proxima 7.71 ns

L3 (M8M) -- Physics BSM

$ds^2 = -c^2 dt^2 + Z^2(r)[dr^2 + r^2 d\Omega^2]$ (Phase-Z metric)
 $Z(r)$: $Z_{vac}=15$ (outside) -> $Z_{throat}=1$ (at throat)
 $G_{eff}(Z) = G_0 * (Z_{vac}/Z_{throat})^4 = 50625 * G_0$

L4 (M8N) -- EEQC v14

$P_L = P_{phys}^d$ where $d=6$ (D20 code distance)
 $P_L \rightarrow 0$ as $P_{phys} < \text{threshold}$
L1 $f_{res}=\alpha_0$ | L2 $Z=15$ | L3 D20 $d=6$ | L4 $\text{tidal}=0.0999g$

L5 (M8O) -- Fault-Tolerant Gates

$G_{eff} = 50625 * G_0$ (Z^4 amplification at $Z_{vac}=15$)
 $P_{hold} = 1.40 \text{ kW}$ (exotic matter hold power)
Inject error $Z=1.002$: ABORT triggered [PASS]

L6 (M8P) -- Logical Clock

$B_M = \alpha_0 / 13.75 = 21.7683024920261 \text{ MHz}$
 $H4_{ratio} = \Omega/R / 11 = 12/11$ (BSD connection)
 $\text{Tr}(\omega) = 0$ (CONTACT ZERO) | $M^* = 4/55$ exact

L7 (M8Q) -- System

Redundancy = 7: system fails iff all 7 layers abort simultaneously
35/35 routes GREEN | 120/120 cells | 1680/1680 PLLs | MTBF=5.5yr
Universal 7-abort matrix: min 7 simultaneous failures to break

SECTION IV -- MORNING STAR HEALTH STATE 6

Six Certification Axes -- all must be GREEN for operational status:

Health State	Axis	Value / Bound	State
HS1 -- Frequency lock	B_M oscillator	21.7683024920261 MHz exact	PASS
HS2 -- FTL advance	v_g / c	3.183 RTT = 18.635 ns	PASS
HS3 -- Geometry	D20 / Z-lock	120 cells Z=15 exact	PASS
HS4 -- Gravity control	G_eff / tidal	50625*G_0 tidal=0.0999g<0.1g	PASS
HS5 -- Quantum coherence	P_logical / 12/11	P_L=0 12/11 handshake	PASS
HS6 -- System integrity	Routes / MTBF	35/35 GREEN MTBF=5.5 yr	PASS

MORNING STAR HEALTH STATE 6: GREEN^6 -- ALL PASS

SECTION V -- NUMERICAL VERIFICATION

Check	Expected	Computed	Status
B_M err vs alpha_0/13.75	<0.5%	0.0000%	PASS
G_eff = 15^4 = 50625	50625	50625	PASS
PLLs: 14*120=1680	1680	1680	PASS
RTT channel geometry	PASS	8.89 m half-length	PASS
B_M = alpha_0/13.75	exact	21.7683024920261 MHz	PASS

System Health Matrix (M8Q):

Parameter	Value	Status
Routes	35/35	GREEN
D20 cells	120/120	PASS
PLLs	1680/1680	PASS
Tidal	<0.1g	PASS
P_logical	0	PASS
MTBF	5.5 years	PASS
EEQC layers	7/7	GREEN^7

Safety Certification (M8I / M8J):

Morris-Thorne wormhole: r0=3m | b'(r0)=0 PASS | E_cav=1.44MWh
Recalibrated (M8J): delta=1.89m | f2=3.21e17 J/m | tidal=0.0999g PASS
Delta_tau=7.647ns (OQ-2 closed PASS) | FTL Cert: MS-FTL-20260523-001

CERTIFICATION SUMMARY

Certification Check	Result
S1 - Parent Module SHAs (M8K,M8L,M8M,M8N,M8O,M8P,M8Q)	PASS
S3a- B_M err < 0.5% vs alpha_0/13.75	PASS
S3b- G_eff = 15^4 = 50625 (exact)	PASS
S3c- PLLs: 14*120=1680 (exact)	PASS
S4 - System health matrix GREEN	PASS
S5 - 7-layer equations documented	PASS
S6 - Morning Star Health State 6: GREEN^6	PASS

MS Tower stdout SHA-256:
86834fbdba0358b0bfff1d4665928986a6c426a86f7ac5b35c416af513838e4cc
Parent module SHAs (M8K through M8Q): see invariants.json

STATUS: MS_TOWER_CERTIFIED

B_M=21.7683024920261MHz | RTT=18.635ns | v_g=3.183c | GREEN^7
35/35 routes | 120/120 cells | 1680/1680 PLLs | MTBF=5.5yr | P_L=0
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